

Slide	Pass	Knob
1	Pass A	Empirical (no sim knob)
2	Pass B	λ in score
3	Pass C	ρ in assignment

Headline
Alex is pleased. You’re not off track. The Pass work is **right**. The shift is **directedness**: fewer items, clearer hierarchy, outputs as a **characterization slide series** — and a hard line between **sensitivity analysis** (now) and **statistical parameter fitting** (next big insight).
His closing: *“This is still basically what I asked for two weeks ago... we need to be more directed.”*

What landed well
Pass C / ρ : Assortativity answered cleanly — flat without ρ , roster/peer effect appears with ρ . He called that “really good.”
Pass B / λ : Directionally right ($\lambda = 0$ vs $\lambda > 0$ bends the curve), but **incomplete** — needs the same systematic treatment as ρ , plus the double-plot ask below.
Pass A / empirical: The real-data side-by-side (talent vs poolq_loo) is the anchor. The “two x-axes” conversation is about making the **generative** slides show the same dual readout the empirical slide already implies.
Model simplicity: Collapsed **L_C** is fine for this paper. B vs D decomposition is parked — development benefits of good teams are real but not identifiable in promotion data yet → one black-box **L**.
Sweep infrastructure: Rivanna/sweep code is good; reshape output into a **standard slide series** (equations at top, bold the knob, figures below) — not one-off PNGs.

Three layers (don’t merge them)
Layer 1 — Gallery fixes (build on what you have)
Concrete asks from looking at the deck:
1. **λ ablation panel** — mirror Pass C’s ρ panel: **fix ρ** , vary λ (e.g. four subplots at four λ values at one fixed ρ , same layout as Pass C).
2. **Double-plot in simulation** — one run (e.g. $\lambda = 0.55$, ρ fixed):
 ◦ **Left**: 16 bins on **individual A_i** (sort 1→n, then bin)
 ◦ **Right**: 16 bins on **poolq_loo** (team average talent, LOO)
 ◦ Same scenario both sides — not separate model runs with different λ .
3. **Overlay $\lambda = 0$ vs $\lambda > 0$** on the **A_i -binned** side for deltas.
4. **Put sort-and-chop back** on the Pass C ρ figure (you cut it because it squished the panel).
5. **Report fitted inputs**: distribution of **t_j , σ^2** , etc. — notation that **data-fitted** quantities ($\hat{t}$, empirical draws) differ from **simulation-drawn** ones.

Alex called these “puzzle pieces” — important, but not the deepest insight. Phase A (below) is the quick gallery pass; Phase B is the same logic turned into a full one-slide-per-knob characterization deck.

Layer 2 — Main mission now: sensitivity characterization (~4 days)
Vary **one knob at a time**; keep everything else fixed; show how the **two curves** move.
The two curves (throughout the meeting):
• binned by **A_i** → monotone talent ladder
• binned by **poolq_loo / L_C** → inverted-U / roster-pressure shape

OAT caveat (interactions): One knob alone might never produce the Hero-style curve; ρ and λ (for example) may be necessary together. Alex’s one-at-a-time plan maps each knob around a **working baseline** (539 preset)—not rediscover the inverted-U from zero. If ρ or λ marginal sweeps look flat, check baseline and consider a small **$\rho \times \lambda$** panel at the checkpoint with Alex; Pass A remains the empirical anchor.

Knobs to characterize (one slide each: 3 core equations, **bold the knob**, figure below):

Knob	Alex’s expectation
ρ	Matters a lot (Pass C shows this)
λ	Matters a lot (Pass B starts it; needs full panel)
θ (sigmoid center)	Matters; likely tied to K/N (success rate), not “median of selected”
γ (sigmoid slope)	Characterize
Distribution of A_i	Probably doesn’t matter much (CLT on team averages) — but document choice + SI robustness
Distribution of T_j	Fix from empirical ballpark; characterize
K/N (selection rate)	System feature — “everybody gets a trophy” vs highly selective regimes

Not yet: full statistical fitting to real data. At ~41:44: *“I’m not caring about fitting to data yet — just characterize what’s going on.”*
Pace: ~two knobs per day → ~four days for a solid characterization deck. **Checkpoint with Alex after the first two** (ρ and λ).

Layer 3 — Next big insight (after characterization): statistical fitting, not curve matching

Do NOT: pick ρ or λ by minimizing distance between empirical Hero curve and simulated curve $\rightarrow$ that's **curve fitting**.

DO: define a **statistic of the raw data** for each parameter:

- ρ : e.g. **within-team correlation of ability** (maybe log-transformed) — model-invariant, computable from data alone
- λ : a **roster-pressure statistic** — autocorrelation / "if a teammate gets promoted, what does that tell you about your odds?" — **not** the sim selection-score formula

If you nail that, you're "putting a metric on the effect" and "really going somewhere."

PD14 magnitude (Model A vs B predictive comparison) is **related** but PD15 is sharper: estimators tied to **parameters**, not just better prediction. **Park magnitude until characterization is done** unless Alex says otherwise.

Other model notes from the conversation

- **B/D vs L**: Use $\mathbf{L}_C$ only for now; future work could split λ_B and λ_D .
- **Notation**: σ used twice (team σ_j vs sigmoid σ) — fix. θ = sigmoid center (not median of selected). γ = sigmoid slope.
- **θ and K/N**: Centering θ likely depends on **K/N** (draft slots / population), a fixed **system property** (MBB vs Army differ) — sweep it, don't collapse it.
- **A_i distribution**: Alex expects **robustness** (enough spread; CLT on team means), not power-law dependence — show alternatives in SI.
- **Process**: 57-item checklist is too many; same failure mode as cranking calculations that feel like progress but aren't the mission.

How this supersedes the old checklist

CHARLES_CHECKLIST.md was re-entry + run Pass A/B + PD14 magnitude. That scaffold did its job.

PD15 says **archive it** and replace with something shorter, phased:

Phase A — Gallery honesty (1–2 days)

λ panel, sim double-plots, sort-and-chop back, fitted-distribution callouts.

Phase B — Characterization deck (~4 days)

$\rho \rightarrow \lambda \rightarrow \theta/\gamma \rightarrow A$ distribution $\rightarrow t_j \rightarrow K/N$. Checkpoint after $\rho + \lambda$.

Phase C — Statistical fitting (later)

Data statistics for $\hat{\rho}$ and $\hat{\lambda}$; then revisit PD14 magnitude.

Still parked: B/D split, Rivanna hero rebuild, army figures, bin-16 prose polish.

Open questions (Charles, 2026-08-03)

1. **Board vs screen**: When Alex said "flat without assortativity," was that **Pass C on screen? Yes — Pass C**.
2. **Double-plot target**: Left = A_i -binned, right = poolq_loo-binned, same sim run? **Yes — confirmed**.
3. **λ sign on Model slide**: Use $S_i = A_i - \lambda L_C$ ($\lambda > 0$; congestion enters with a minus sign). Matches the meeting correction to Alex.
4. **PD14**: Park until after Phase B, or keep as parallel track? **Park until after Phase B**.

Bottom line

You didn't waste the last two weeks. Alex said the Pass work is **right**. The job now is **directed characterization** — fewer checklist items, standard slides, two curves every time — then **fit ρ and λ to data statistics**, not to curve shape.

When you want, we can archive CHARLES_CHECKLIST.md and draft a **~15–20 item PD15 checklist** using **Pass A / B / C as on your slides**.